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Project Title:	Induction Of Bystander Effects By High LET Radiation In Cells And In Tissue Models		
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The mission of NASA’s space radiation biology program is to provide knowledge on the effects of high energy charged particles on biological systems at the molecular, cellular, tissue and organism levels in order to enable better prediction of risks of carcinogenesis and damage to normal tissues from exposures of astronauts to the ionizing radiation in space. With charged particles such as alpha-particles and low energy protons, significant differences from direct DNA damage-mediated responses have been demonstrated in a number of biological systems. Occurrence of these effects, including bystander effects and genomic instability, calls into question the use of a simple linear extrapolation of cancer risks from high dose data to low doses. This is of particular relevance to NASA because the occurrence of bystander effects in cells exposed to heavy ions or energetic protons, as found in space, has not been evaluated to date. The overall goals of this project are to test the hypothesis that heavy ions and energetic protons efficiently cause bystander effects in neighboring cells not directly traversed by the radiation and to study the mechanisms involved. There are four specific

Task Description:	aims: 1. Determine induction of bystander effects by heavy ions compared to X-rays and protons in cultured human diploid fibroblasts and keratinocytes using endpoints specific for each cell type and including production of micronuclei, apoptosis, clonogenicity and increased gene expression. 2. Test the hypothesis that heavy ion-induced bystander effects can be modified by exposure of cells to reactive oxygen species (ROS), an overlying oxidative stress. 3. Test the hypothesis that exposure to low LET radiation alters the heavy ion-induced bystander effects. 4. Begin development of 3-D tissue models for assessment of bystander effects from heavy ions and energetic protons. The studies specifically address the Research Emphasis for FY2003 of “Molecular radiation biology of carcinogenesis”, as described in the NASA Research Announcement. The studies will aid in assessing the risks to astronauts of carcinogenesis as a result of exposures to radiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research should provide added insight into mechanisms involved in radiation-induced carcinogenesis and cell damage, particularly at low doses. Such information is highly relevant to determination of the shape of the radiation dose response curves for cancer induction or other late tissue damages at low doses of radiation; these dose responses are used to predict risk of damages from radiation. The work with heavy ions and protons, which are components of cosmic rays, is particularly relevant for risk assessment for astronauts. However, the information gained also could be relevant to risks for induction of second cancers in patients treated with radiation therapy or to radiation workers or the general public exposed to low doses of ionizing radiation (e.g., from radon).
Task Progress:	We have developed a novel transwell insert co-culture system in which irradiated and unirradiated cells share medium, but are physically separated, and used this approach to demonstrate medium-mediated bystander responses in AG01522 normal human fibroblasts after X-ray exposure (Yang et al. 2005). At the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory we have used this co-culture system, as well as irradiating cells at low particle fluences, to study bystander responses induced by heavy ions (1 GeV/n Fe, 1 GeV/n Ti and 290 MeV/n C) and protons (1 GeV/n). We have used iron, titanium, carbon and protons because they represent particles found in space that span a range of LET values from 0.24 to about 150 keV/micrometer, thus allowing us to test the LET dependence of various bystander responses. We have found substantial effects in directly irradiated AG01522 fibroblasts and h-TERT immortalized human keratinocytes using endpoints including clonogenic survival, micronuclei (MN) formation and p21 expression. For example, with the clonogenic survival assay in the AG01522 cells, we have found relative biological effect (RBE) values at 10% survival are 2.5, 2.3, 1.0 and 1.2 for 1 GeV/amu Fe, 1 GeV/amu Ti, 290 MeV/amu C and 1 GeV/amu protons, respectively, compared to 250 kVp X-rays, indicating the greater effectiveness of Fe and Ti for inducing cell killing. Fe and Ti are also very effective inducers of MN in these cells, although C ions are less effective, similar to protons. These results were reported this year (Yang et al. 2007a). Using our insert system, we have demonstrated that irradiation with 1 GeV/amu Fe ions induces a 2-3 fold increase in MN formation in unirradiated bystander AG0 cells, which is similar in magnitude to that seen with X-rays. The increase in damage in bystander cells is independent of the dose to the hit cells between 0.1 and 2 Gy Fe ions, although it is somewhat lower at 0.05 Gy. The insert system allows us to study the timing of bystander signaling by altering the time at which the co-culture of irradiated and unirradiated cells is started or how long the two cell populations are together. Using this approach, we have shown that when irradiated and unirradiated bystander cells are combined in shared medium immediately, 1 h or 3 h after irradiation, a full bystander MN formation effect is observed. However, when combining the irradiated and unirradiated cells is delayed 6 h after irradiation, the bystander effect is lost. Furthermore, bystander micronuclei formation is observed even when the bystander cells are co-cultured with irradiated cells for only 1 h if they are combined immediately after irradiation. This suggests that the signal transduction to bystander cells from irradiated cells occurs at a short time after irradiation, and these early stage signals sent by irradiated cells are crucial to the induction of bystander effects. The results with iron ions are similar to those with X-rays, suggesting that the bystander MN formation in this system is independent of LET (Yang et al. 2007b). Protons also induce a medium-mediated bystander effect in terms of MN production, although the magnitude is less than seen with Fe ions or X-rays. We were the first to report gamma-H2AX foci formation in bystander cells (Yang et al. 2005). In addition to X-irradiation, we have now shown gamma-H2AX foci formation in unirradiated bystander cells that shared medium with Fe ion-irradiated cells at different time points after co-culture. Our data indicate that a 2-3-fold increase in percentage of bystander cells with gamma-H2AX foci occurs as early as one hour after irradiation, and lasts at least 24 h. Compared to the directly irradiated cells, which have a high yield of gamma-H2AX foci at early times after irradiation and return to control levels by 24 h, bystander cells have slower kinetics of disappearance of gamma-H2AX foci. Fe ions and X-rays induce a similar pattern of gamma-H2AX foci in unirradiated bystander cells, suggesting gamma-H2AX foci induction in bystander cells is independent of LET of the radiation (Yang et al. 2007b). We have been interested in investigating effects of very low fluences of particles where only 1-2% or less of the cells would be traversed by particles. Dr. Rusek and his team of NSRL physicists have developed procedures for obtaining these very low fluences, and, most importantly, being able to accurately quantify the particle fluences, determine the beam uniformity and determine the purity of the beam. We have now demonstrated that very low fluences of energetic protons and iron ions induce bystander MN formation and 53BP1 foci (another surrogate for DNA DSBs) in non-irradiated bystander fibroblasts that share medium with irradiated cells. Our results show that while there is no significant increase in MN formation in cells directly irradiated with low fluences (2e3 to 5e6 particles/cm2 for protons and 2e3 to 2.25e5 particles/cm2 for iron ions) of protons and iron ions, there is 1.5-2-fold increase in MN formation in bystander cells in the insert system. On the other hand, 53BP1 forms foci in both directly irradiated and bystander cells with low fluences of protons and iron ion, with the percentage of irradiated cells with 53BP1 foci being a little higher than that of bystander cells. It is noteworthy that within the range of low fluences we used, 53BP1 foci formation is independent of dose. This suggests that the linear no-threshold hypothesis for relationship of radiation-induced damage to dose may not hold at very low doses. Furthermore, although we observe the typical dose response plateau pattern of irradiation-induced bystander effects, the fluence at which the threshold occurs is still unclear. At 2e3 particles/cm2, which equals 0.4 mGy of iron ions or 0.57 microGy of protons, there is bystander MN formation with iron ion irradiation

	but not with protons. The lowest proton fluence that induces the bystander effect is 2e5 particles/cm2, which equals 57 microGy. These data suggest that there might be a threshold for irradiation-induced bystander effects at about 60 microGy for protons and somewhere below 400 microGy for Fe ions. With regard to aim 2 we have reported previously that ROS play a significant role in transmission of signals from irradiated to bystander cells after exposure to X-rays or Fe ions (Yang et al., 2005, 2007b). We have now begun studies to determine the effect of low levels of oxidative stress on responses of irradiated and bystander cells to subsequent irradiation. Our preliminary data show that if bystander AG0 cells are exposed to a low, non-toxic level of hydrogen peroxide before co-culturing with irradiated cells, there does not appear to be any radiation-induced bystander MN formation, but if bystander cells are oxidatively stressed after the start of co-culturing with irradiated cells, the radiation-induced bystander MN formation is not changed compared to the sham control. Further studies are needed to confirm those results. Relevant to aim 3, we have been among the pioneering groups at NSRL to investigate the effects of “mixed”, actually sequential, irradiations with iron ions and protons. Our preliminary data suggest that in irradiated cells the induction of micronuclei is greater when the two particle exposures are separated by 3 or 24 h compared to when given within 5 minutes. However, depending on the time between irradiations, bystander cells appear to “adapt” to the second signal, i.e., a smaller bystander response was seen when irradiated cells were “hit” twice with 3-24 h between the hits, than when the same dose was given in a single irradiation. Clearly, these data need repeated, but the results are intriguing. Previously, we reported progress on aim 4 showing significant signaling from X-irradiated fibroblasts to unirradiated keratinocytes and vice versa (endpoints of MN formation and p21 expression). However, interestingly, after Fe exposure our data suggest that with some endpoints there is no signaling from irradiated keratinocytes to unirradiated fibroblasts or keratinocytes, and the effect is delayed from irradiated fibroblasts to unirradiated keratinocytes. Additional investigation is still underway, and we have made progress developing a 3D tissue model using these two cell types that will allow us to test the importance of microenvironment to cell-cell signaling. In summary, in the past four years we have made significant progress on our aims for this project and have obtained a number of interesting, novel findings in both directly irradiated and bystander cells. These findings have been published in different journals.
Bibliography Type:	Description: (Last Updated: 11/13/2018)
Abstracts for Journals and Proceedings	Held KD, Yang H, Anzenberg V. "Bystander effects induced by heavy ions and energetic protons." Presented at the 4th International Workshop on Space Radiation and 17th Annual NASA Space Radiation Investigators' Workshop, Moscow and St. Petersburg, Russia, June 2006. Proceedings of the 4th International Workshop on Space Radiation and 17th Annual NASA Space Radiation Investigators' Workshop, Moscow and St. Petersburg, Russia, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Held KD, Yang H, Purshcke M, Spantchak Y. "Bystander effects induced by low fluences of energetic protons and iron ions in human normal fibroblasts." Presented at the 18th Annual NASA Space Radiation Investigators' Workshop, Sonoma, CA, July 2007. Proceedings of the 18th Annual NASA Space Radiation Investigators' Workshop, Sonoma, CA, July 2007. , Jul-2007
Articles in Peer-reviewed Journals	Yang H, Anzenberg V, Held KD. "The time dependence of bystander responses induced by iron ion radiation in normal human skin fibroblasts." Radiat Res. 2007 Sep;168(3):292-8. PMID: 17705636 , Sep-2007
Articles in Peer-reviewed Journals	Yang H, Anzenberg V, Held KD. "Effects of heavy ions and energetic protons on normal human fibroblasts." Radiats Biol Radioecol. 2007 May-Jun;47(3):302-6. PMID: 17867499 , May-2007
Books/Book Chapters	Yang H, Held KD. "Radiation-induced bystander effects in cancer." in "Reviews in Cancer Biology & Therapeutics." Ed. U.N. Kasid, V. Notario, A. Haimovitz-Friedman, M. Bar-Eli. Kerala, India : Transworld Research Network, 2007., Dec-2007